CSSE497 Standards Assignment (with XP)

The Institute's accrediting body, ABET, expects that you engage with the *standards* of professional bodies such as IEEE, ISO, etc. We would especially like you to think about how a particularly broad standard, **IEEE 12207-2017**, relates to the Extreme Programming practices that we apply to our projects.

However, IEEE's standards are entirely opt-in and completely disclaim all liability from their use in your project. Those caveats don't inspire confidence in the application of the standards. So, we expect you to **think critically** about whether IEEE's standards are applicable and/or useful to you. This thought process (everything has a process in standards-land!) is called *tailoring the standard* by IEEE. These instructions will walk you through the tailoring process to produce a Tailored Engineering Standard for Senior Project.

Individual Task – Complete the Logan Library's Standards training online

Our library has an online training module that tells you all about standards and why they are purported to be important to some engineering fields. See CSSE497 Moodle for the link.

Complete the linked standards training on Moodle before you continue to the next step.

Group Task 2- Tailor the IEEE standards to your project

Only start this after everyone has completed the Individual Task.

In your git repo, create a new markdown file named:
Tailored Engineering Standards for Senior Project.md.

Go to the library and ask to check out standard **IEEE 12207-2017** from the CSSE course reserves. [do not wait until 10th week to make this request]
Specifically, read each of sections of the standard, which form the core Software-Specific Processes:

Call #
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12207

- 6.4.2 Stakeholder Needs and Requirements Definition process
- 6.4.4 Architecture Definition process
- 6.4.5 Design Definition process

Note: Each of these sections is ~4 pages in length, the entire document ~140 pages

Tailoring each of the Processes

For each of the Processes listed above:

1. Create a new heading in your markdown document for the Process you are reading. In markdown syntax, your first heading should be written as:

   ```
   # 6.4.2 Stakeholder Needs and Requirements Definition process
   ```

2. Read the **Purpose** and **Outcomes** sub-sections of the Process. These tell you what the IEEE wish would be true of your process.

3. Read the **Activities and tasks** sub-section of the Process. It tells you what the IEEE wishes you would do, generally, to achieve their stated outcomes.
4. Make sub-headings for each activity found in the Activities section. The first one should be written in markdown as:

   ```
   ## 1. Prepare for Stakeholder Needs and Requirements Definition
   ```

5. Each activity is broken down into tasks, so create sub-sub-headings for each Task associated with that activity. The first heading should be written in markdown on one line as:

   ```
   ### a. Identify the stakeholders who have an interest in the
   software system throughout its lifecycle.
   ```

6. For each Task, answering the following questions:
   a. What Outcome of the Process does this task seem to address? The IEEE document lists each outcome using a unique letter (e.g., 'a', 'b', 'c', etc.). For your response to this part, just provide the single letter designator.
   b. What is the Extreme Programming implementation of this task? Provide direct quotes from the XP book to support your answer.

Repeat these steps for **each** of the three listed Processes (above).

In summary: the standards belong to us

We examined IEEE 12207 as just one example of an applicable standard. There are **many** more standards covering **all** aspects of software development. (See if you can find the ones that discuss how to do XP properly!) Regardless of the problem you are trying to solve, the standards probably have an OK process for outlining considerations to make. If there is something you don't like about a standard, then you can always tailor it, like we did here. The point is to be intentional and thorough about your engagement with standards.

The standards are written by the collective wisdom of our professional societies and are updated regularly as the times change. If you categorically don't like our field's standards, then join our professional societies, and shape the future of the standards yourself.

Submission instructions

1. Complete the training on the library's website.
2. When you are done responding to all listed Processes: turn in this Assignment to the Moodle folder (one person per team).